

EXPERIMENT 38

Basic Configuration Setup for Installing Hadoop 2.x — Creating HDUSER and SSH Localhost

AIM

To perform the basic configuration setup required prior to installing Hadoop 2.x, including creation of a dedicated Hadoop group and user (hduser) and configuration of passwordless SSH access to localhost.

REQUIREMENTS

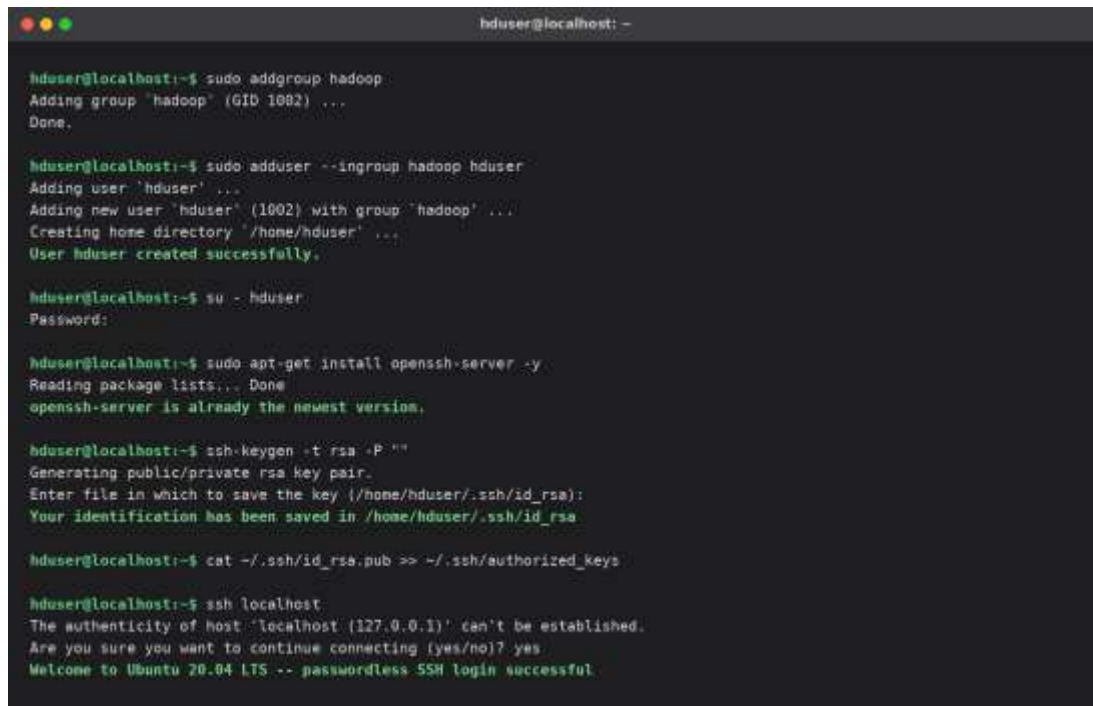
- Ubuntu Linux system (physical or virtual machine)
- Terminal access with sudo privileges
- OpenSSH server and client packages
- Java Development Kit (prerequisite for Hadoop)

PROCEDURE

1. Open a terminal and create a dedicated group named hadoop using the addgroup command.
2. Create a new user hduser and add it to the hadoop group using the adduser command.
3. Switch to the hduser account using the su command.
4. Install the OpenSSH server package if it is not already installed.
5. Generate an RSA SSH key pair for hduser with an empty passphrase using ssh-keygen.
6. Append the generated public key to the authorized_keys file to enable passwordless login.
7. Set appropriate permissions on the .ssh directory and authorized_keys file.
8. Test the configuration by connecting to localhost using the ssh command.
9. Confirm the host key and verify that login to localhost succeeds without a password prompt.

IMPLEMENTATION

The hadoop group and hduser account are created first so that all Hadoop services run under a dedicated, isolated user. Passwordless SSH to localhost is then configured, since Hadoop's start/stop scripts use SSH internally to manage daemons even on a single-node (pseudo-distributed) setup.



```
hduser@localhost: ~  
  
hduser@localhost:~$ sudo addgroup hadoop  
Adding group 'hadoop' (GID 1002) ...  
Done.  
  
hduser@localhost:~$ sudo adduser --ingroup hadoop hduser  
Adding user 'hduser' ...  
Adding new user 'hduser' (1002) with group 'hadoop' ...  
Creating home directory '/home/hduser' ...  
User 'hduser' created successfully.  
  
hduser@localhost:~$ su - hduser  
Password:  
  
hduser@localhost:~$ sudo apt-get install openssh-server -y  
Reading package lists... Done  
openssh-server is already the newest version.  
  
hduser@localhost:~$ ssh-keygen -t rsa -P ""  
Generating public/private rsa key pair.  
Enter file in which to save the key (/home/hduser/.ssh/id_rsa):  
Your identification has been saved in /home/hduser/.ssh/id_rsa  
  
hduser@localhost:~$ cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys  
  
hduser@localhost:~$ ssh localhost  
The authenticity of host 'localhost (127.0.0.1)' can't be established.  
Are you sure you want to continue connecting (yes/no)? yes  
Welcome to Ubuntu 20.04 LTS -- passwordless SSH login successful
```

Figure 1: Illustrative terminal view — hduser creation and passwordless SSH configuration.

RESULT

The hadoop group and hduser account were created successfully, and passwordless SSH login to localhost was configured and verified for hduser.